1. Change Request Details

Field	Details
Change Request ID	CR-019
Change Request Title	Student Profile Editing
Student Name	S.Hindusha
Student ID	JF/ICT/24/26
Implementation Status	Completed
Date	29 October 2026

2. Affected Files

File	Change	Purpose
backend/src/controllers/studentController.js	Modified	Implements the student profile update controller and validation logic.
Backend/src/models/studentModel.js	Modified	Adds the database operation for updating student contact information.
Backend/src/routes/studentRoutes.js	Modified	Registers the protected profile update endpoint.
Frontend/src/pages/StudentDashboard.jsx	Modified, if applicable	Provides access to the student profile editing interface.
Frontend profile component or form	Modified, if applicable	Allows students to edit and submit their email address and phone number.

3. Testing and Verification

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Log in using valid student credentials.	Student login succeeds.	Student login succeeded.	Pass
TC-02	Open the student profile section.	Profile opens successfully.	Profile opened successfully.	Pass
TC-03	View existing email and phone number.	Profile information is displayed correctly.	Profile information was displayed.	Pass
TC-04	Update email with a valid address.	Email is updated successfully.	Email was updated successfully.	Pass
TC-05	Update phone with a valid ten-digit number.	Phone number is updated successfully.	Phone number was updated successfully.	Pass
TC-06	Update both email and phone number.	Both values are saved successfully.	Both values were updated successfully.	Pass
TC-07	Verify profile after saving.	Updated values are displayed.	Updated values were displayed.	Pass
TC-08	Submit an invalid email address.	Invalid email is rejected.	Invalid email validation was verified.	Pass
TC-09	Submit an empty email address.	Request is rejected with a validation message.	Empty email validation was verified.	Pass
TC-10	Submit a phone number containing letters.	Invalid phone number is rejected.	Phone number validation was verified.	Pass
TC-11	Submit a phone number with fewer than ten digits.	Phone number is rejected.	Length validation was verified.	Pass
TC-12	Submit a phone number with more than ten digits.	Phone number is rejected.	Length validation was verified.	Pass

TC-13	Submit an email already used by another student.	Duplicate email is rejected.	Duplicate email validation was verified.	Pass
TC-14	Submit a request without an authentication token.	Request is rejected.	Authentication protection was verified.	Pass
TC-15	Submit an invalid or expired token.	Request is rejected.	Invalid token handling was verified.	Pass
TC-16	Access the endpoint using an unauthorized role.	Request is rejected.	Role-based authorization was verified.	Pass
TC-17	Attempt to update another student's profile.	Other student's profile remains unchanged.	Ownership protection was verified.	Pass
TC-18	Verify the registration number after updating.	Registration number remains unchanged.	Registration number remained unchanged.	Pass
TC-19	Verify the student's name after updating.	Student's name remains unchanged.	Student's name remained unchanged.	Pass
TC-20	Reopen the profile after updating.	Updated information persists.	Updated information persisted.	Pass
TC-21	Verify frontend API error handling.	Appropriate error message is displayed.	Error handling was verified.	Pass
TC-22	Verify student login and profile viewing.	Existing functions continue to work.	Existing functionality was verified.	Pass
TC-23	Verify the password change function.	Password change continues to work.	Password change was verified.	Pass
TC-24	Verify administrator student management.	Administrator functions remain operational.	Administrator functionality was verified.	Pass

4. Test Summary

Test Result	Count
Total Test Cases	24
Passed	24
Failed	0
Pass Rate	100%
Overall Result	Pass

Use these results only if all 24 test cases were actually executed and verified.

5. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Student can view their profile.	Frontend verification	Met
AC-002	Student can update their email address.	Profile update test	Met
AC-003	Student can update their phone number.	Profile update test	Met
AC-004	Backend provides a profile update endpoint.	API testing	Met
AC-005	Endpoint requires authentication.	Authentication test	Met
AC-006	Endpoint enforces the required student role.	Authorization test	Met
AC-007	Student identity is obtained from authenticated request information.	Controller and API verification	Met
AC-008	Invalid email addresses are rejected.	Validation test	Met

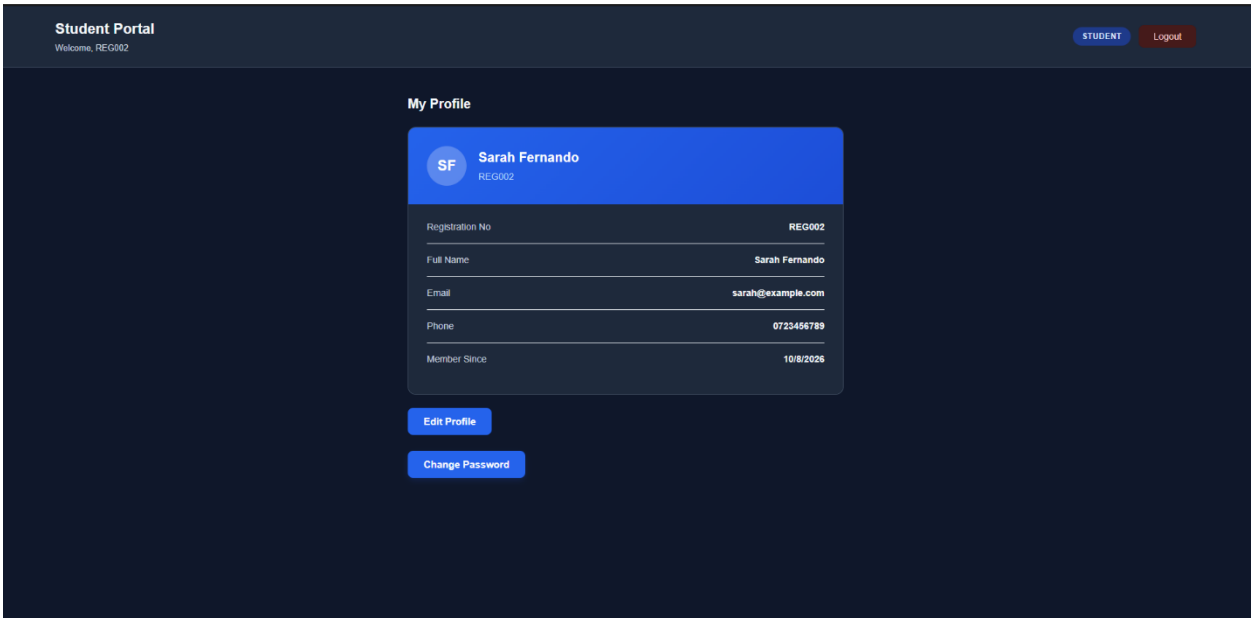
AC-009	Invalid phone numbers are rejected.	Validation test	Met
AC-010	Duplicate email addresses are rejected where required.	Duplicate email test	Met
AC-011	Valid information is saved to the database.	Database verification	Met
AC-012	Updated information is displayed after saving.	Frontend verification	Met
AC-013	Students cannot update other students' profiles.	Authorization test	Met
AC-014	Unrelated student information remains unchanged.	Regression testing	Met
AC-015	Appropriate success and error responses are returned.	API response testing	Met
AC-016	Existing login functionality remains operational.	Regression testing	Met
AC-017	Existing profile viewing remains operational.	Regression testing	Met
AC-018	Existing password change functionality remains operational.	Regression testing	Met
AC-019	Administrator student management remains operational.	Regression testing	Met
AC-020	No unnecessary database schema changes are introduced.	Code and database inspection	Met

6. Evidence Table

Evidence 01 – Student Profile Display

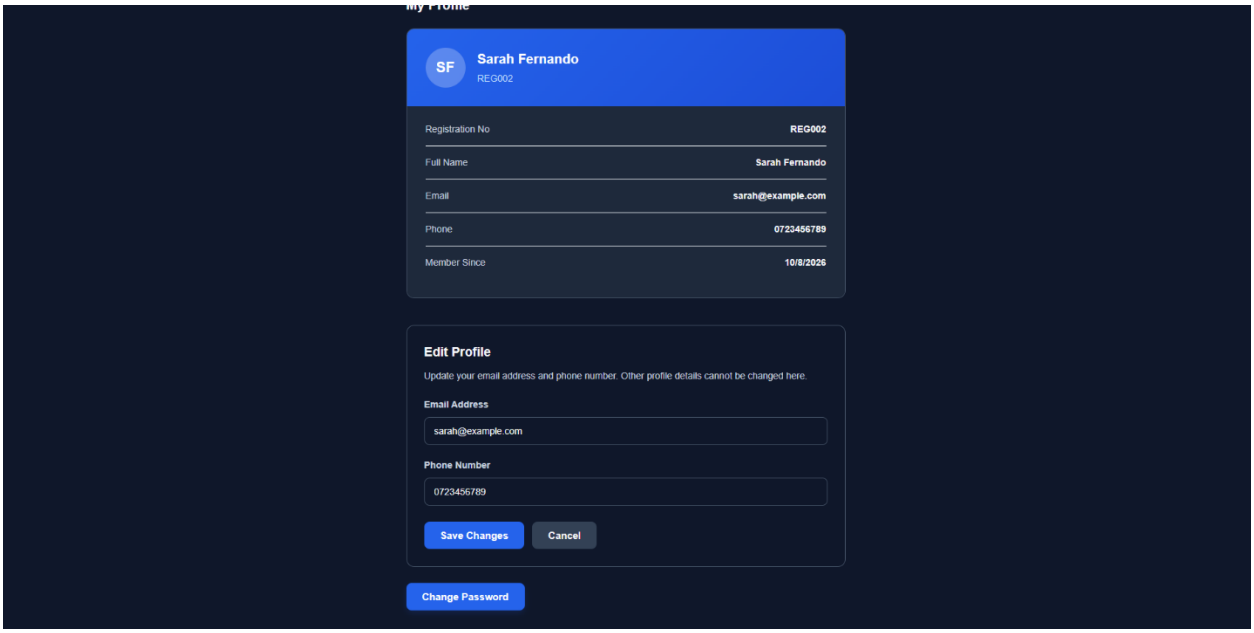
Description: Screenshot showing the existing student profile information.

Related Acceptance Criteria: AC-001



Evidence 02 – Successful Email Update

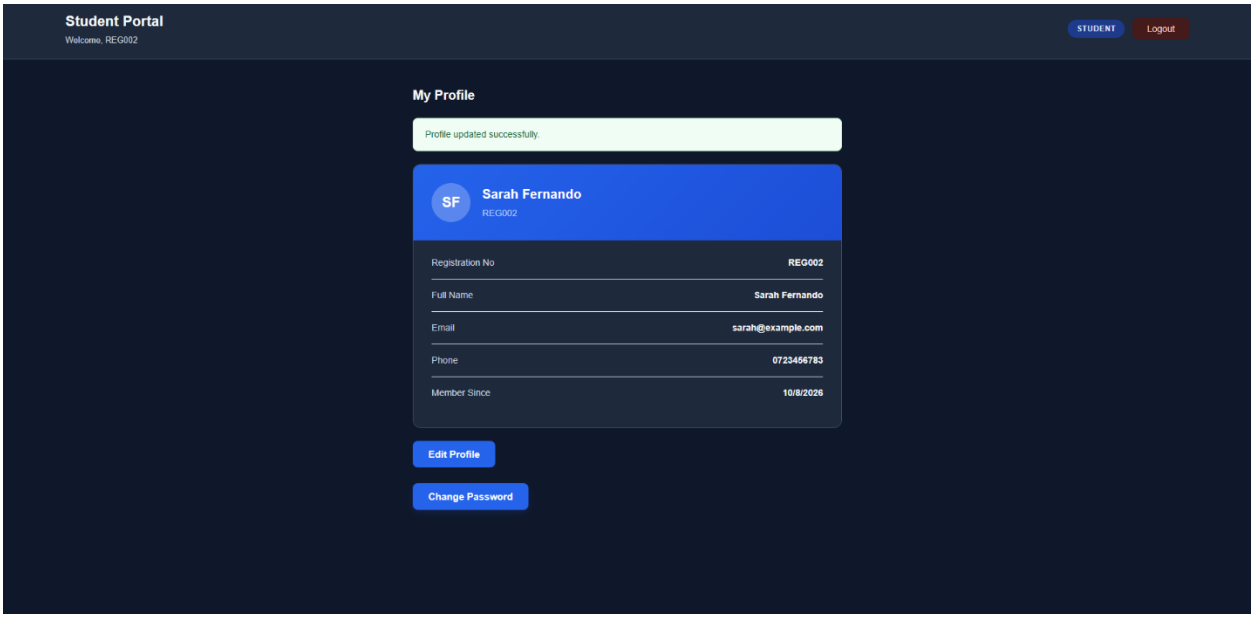
Description: Screenshot showing the updated email address and success message.



Evidence 04 – Updated Profile Information

Description: Screenshot showing the saved email address and phone number after reopening or refreshing the profile.

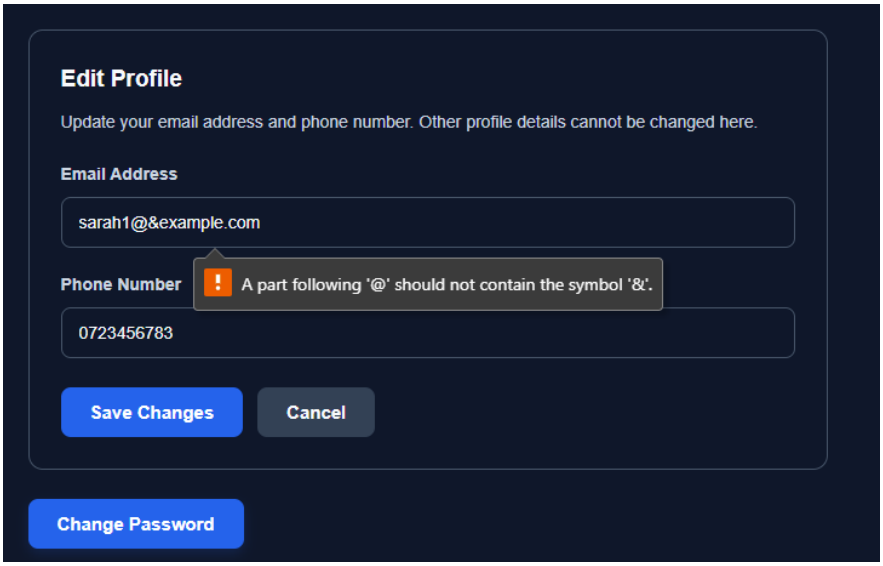
Related Acceptance Criteria: AC-011, AC-012



Evidence 05 – Invalid Email Validation

Description: Screenshot showing the rejection of an invalid email address.

Related Acceptance Criteria: AC-008



Evidence 06 – Invalid Phone Number Validation

Description: Screenshot showing the rejection of an invalid phone number.

